F20FO Digital Forensics Coursework 2

Digital Forensic Investigation on a Seized Virtual Machine

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Contents

Introduction

This report outlines the development and evaluation of a virtual forensic laboratory designed for the testing and recovery of forensic evidence using specialized tools within a Windows-based virtual machine (VM). The Windows VM is configured to operate in complete isolation, with no external network communication, ensuring the integrity of forensic evidence by eliminating any risk of contamination during analysis. It is equipped with industry-standard tools such as FTK Imager, John the Ripper, Hashcat, and HexEdit to conduct comprehensive forensic investigations. Additionally, an Ubuntu-based VM is utilized, also configured in full isolation with no external network access. This VM uses GuyMager to create a second forensic image as part of the coursework.

Forensic Image Process

The forensic Imaging Process is a critical step that involves the creation of an exact bit-for-bit copy of the digital storage (In this case, the VMDK files). This procedure preserves the original data in a forensically sound manner, ensuring the authenticity and integrity of the evidence throughout the investigation

FTK Imager and GuyMager

The first tool involved FTK imager, which is a widely recognized utility for creating and viewing forensic images in a manner that maintains data integrity (Exterro, 2023) and was located on Windows VM. The process first involved downloading the evidence from One Drive to a newly formatted USB to move it to the VM more efficiently and carefully. Once the files are transferred to the formatted USB, it is then loaded onto FTK Imager to create the forensically sound image to analyze further. Within FTK Imager, the selected source type was set to an "image file". This is because the VMDK files were already acquired, and therefore, there was no need to physically copy them again.

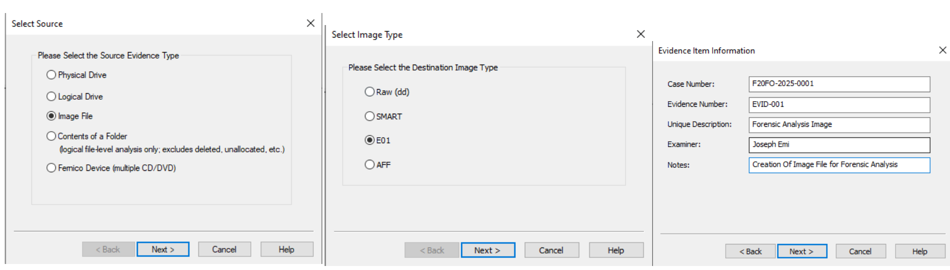

Fig 1.1: FTK imaging Process

Figure 1.1 shows the process of creating the image, from selecting the appropriate image format to giving it a description. The E01 format was chosen due to its forensic advantages, including built-in compression, metadata storage, and integrity verification through embedded hashing.

After completion, the imaging tool generates a computed hash value for the image. This hash is documented and used throughout the forensic analysis process to verify and ensure the integrity and authenticity of the acquired image.

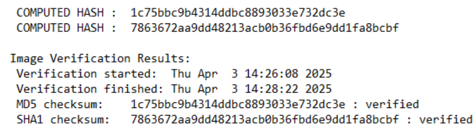

Fig 1.2: FTK image Hash

The next tool is GuyMager, which is forensic imaging tool designed for efficient and secure acquisition of digital evidence on Linux systems (guymager.sourceforge.io, n.d.). The process is very similar to FTK where first, the evidence was transferred from the USB to the Ubuntu VM, where it was then mounted onto the OS using *qemu* for imaging.

Fig 1.3: GuyMager Imaging Process

Figure 1.3 shows a similar process where information regarding the Image, such as format and description, should be filled out. After the process is done, a hash is generated to verify the authenticity and integrity of the image during analysis.

Forensic Analysis

The first forensic analysis was done using the tool Autopsy on Windows VM. After the imaging process, the images were then loaded onto autopsy within Windows to conduct analysis and tag “interesting” files that would be important.

Interesting Files and Recovery

Throughout the analysis process of finding, only certain directories were of keen interest within the disk. The first one is the home directory of the user *f21fo-cw2,* where the user had a lot of interesting files within his documents and pictures directory. Another interesting directory was the */etc* directory, where files of the existing users' passwords were also recovered. Finally, the */var/logs* directory had logs based on what happened to the system before it was seized, which could be of importance to the case. The table below shows the different files that were analyzed and tagged for further inspection:

Notable Files	Types/Method
/img_Image.E01/vol_vol2/home/f21fo-cw2/.bash_history	Info

/img_Image.E01/vol_vol2/home/f21fo-cw2/my-ssh	Info
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/one day in heaven	Extraction/Carving
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/myhiddenfile.txt	Extension Change
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/sys_cooking.dll	Extension Change
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/sys_phare.dll	Extension Change
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/kanban	Extraction/Carving
/img_Image.E01/vol_vol2/home/f21fo-cw2/Downloads/slacker.exe	Extension Change
/img_Image.E01/vol_vol2/home/f21fo-cw2/Downloads/timestomp.exe	Extension Change
/img_Image.E01/vol_vol2/etc/shadow	Info
/img_Image.E01/vol_vol2/var/log/kern.log	Info
/img_Image.E01/vol_vol2/var/log/syslog	Info
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/4028-Astronomers/Albert Einstein.doc	Brute force
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/4028-Astronomers/Nicolaus Copernicus.doc	Brute force
/img_Image.E01/vol_vol2/home/f21fo-cw2/.cache/vmware/drag_and_drop/6Azlf7/2761-Hewish Work[8384].doc	Brute force
/img_Image.E01/vol_vol2/home/f21fo-cw2/Pictures/Agfa_DC-733s_0_936.JPG	Extension Mismatch
/img_Image.E01/vol_vol2/lib/firmware/vxge/X3fw.ncf	Brute force

/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/513-Solar System/Comets	Extension Change

Table 1.5: Notable Files

Informative Files

Many files within the forensic image contained basic but potentially crucial information related to the case. This included the user's bash history, system logs (such as *syslog*), and even the shadow file, which could be brute forced if weak credentials were used. These files, although simple, hold valuable insights and are commonly relevant in most forensic investigations.

File Extension Manipulation

A noticeable tactic within the forensic image was the use of incorrect file extensions, likely used to obscure the files' true nature from casual inspection. Using Autopsy, the tool's file signature analysis proved extremely helpful in identifying these manipulations. For instance, a file named *sys_cooking.dll* was a JPEG image pretending to be a DLL file, hiding in plain sight until revealed during analysis. Another valuable tool utilized during the analysis was HexEdit, which allows forensic investigators to examine raw file data and identify file types based on header values or signature bytes, often called magic numbers (Duttke, n.d.). By looking at the initial byte sequences, it is possible to verify file authenticity or detect mismatches between a file's content and its labeled extension. For example, files pretending to be DLLs but containing headers such as FF D8 FF (indicates a JPEG image) were flagged for further inspection and extraction.

/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/myhiddenfile.txt	Original JPEG
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/sys_cooking.dll	Original JPEG
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/sys_phare.dll	Original JPEG
/img_Image.E01/vol_vol2/home/f21fo-cw2/Downloads/slacker.exe	Original JPEG
/img_Image.E01/vol_vol2/home/f21fo-cw2/Downloads/timestomp.exe	Original JPEG

Fig 1.6: Extension Mismatch

Brute force

Another tactic employed by the suspect involved the encryption of certain files, particularly the .doc documents using password protection. This method restricted direct access to file contents and posed a challenge during the initial analysis. Autopsy, while effective for file recovery and metadata extraction (Carrier, 2023), lacks password-cracking capabilities, presenting a limit in handling encrypted documents.

To overcome this, a two-step approach was adopted using John the Ripper and Hashcat:

- **Password Hash Extraction**: John the Ripper, a widely used password-cracking tool known for its ability to extract and process password hashes from various file types (Openwall, 2022) was used to convert the .doc into hashes. These hashes are then used on another program called Hashcat to crack them based on a given wordlist.

- **Password Cracking:** The extracted hashes were then processed using Hashcat which is a GPU-accelerated recovery tool optimized for speed and capable of performing attacks like dictionary (Hashcat, 2018), A dictionary attack was carried out using the *rockyou.txt* wordlist, while leveraging the power of a GPU. Due to the weak or commonly used passwords, many of the documents were successfully encrypted

This process highlights a key limitation in Autopsy when dealing with password-protected content and demonstrates the importance of having not one tool but many, constantly supplementing forensic workflows with specialized tools for cryptographic analysis.

/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/4028-Astronomers/Albert Einstein.doc	Password: relativity
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/4028-Astronomers/Nicolaus Copernicus.doc	Password: wavelength
/img_Image.E01/vol_vol2/home/f21fo-cw2/.cache/vmware/drag_and_drop/6Azlf7/2761-Hewish Work[8384].doc	Password: heliocentric
/img_Image.E01/vol_vol2/lib/firmware/vxge/X3fw.ncf	Password: unknown

Fig 1.7: Brute force Doc

Extraction and File Carving

Another sophisticated tactic identified during the forensic analysis was the embedding of multiple files within a single file, a method often used to hide data or evade detection. These embedded files were not immediately visible through regular inspection and lacked standard file extensions, making them difficult to identify using Autopsy (Carrier, 2023).

A specific case involved a file titled "one day in heaven", which had no file extension, and an unusually large file size compared to typical plaintext or metadata files. As Autopsy could not extract or analyze embedded file structures, further analysis was conducted using HexEdit. Through manual inspection, distinct file headers (magic numbers) such as FF D8 (JPEG) and 50 4B (ZIP) were discovered at various places, indicating the presence of multiple embedded files (Duttke, n.d.).

To automate the extraction process, the tool Binwalk which is firmware and embedded file analysis tool designed to locate and extract embedded files and executable code within binary images (Devttys0, 2023). This was employed within a Linux environment and scanned the file for known signatures and successfully extracted several concealed files, including one named “my secret.txt”, which contained the suspicious phrase “Funckilicious”. This word could potentially serve as a steganographic passphrase or indicator, suggesting the presence of further hidden content.

Another file, “Kanban”, also lacked an extension and was analyzed using HexEdit. The beginning of the file contained an obfuscated header, which, upon deeper inspection, transitioned into a recognizable JPEG header signature (FF D8 FF). The obfuscation appeared to be a simple technique to hide files from plain sites. The identified header and corresponding data block were manually carved out and reconstructed, revealing a valid JPEG image.

This part of the investigation demonstrates the need for manual validation tools and signature-based carving in forensic workflows, especially when dealing with anti-forensics techniques such as header manipulation and file embedding.

/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/one day in heaven	Files: secret.txt Carved_image1.jpg Carved_image2.jpg
/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/kanban	Files: Carved_image1.jpg
Carved_Image1 (From Kanban)	Carved_sound.wav

Fig 1.8: Carved and Extracted Files

in conclusion, while a single forensic tool can provide useful insights, its limitations often restrict how deep one can go during analysis. Integrating multiple specialized tools allows for a more comprehensive and efficient investigation and this approach also enhances accuracy, adaptability, and the overall integrity of forensic workflows.

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Appendix A: Imaging Pictures

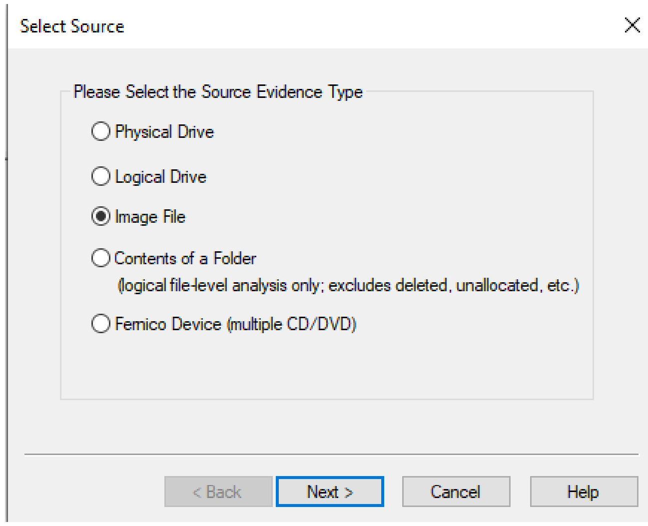

Figure 2.1: FTK Image file option

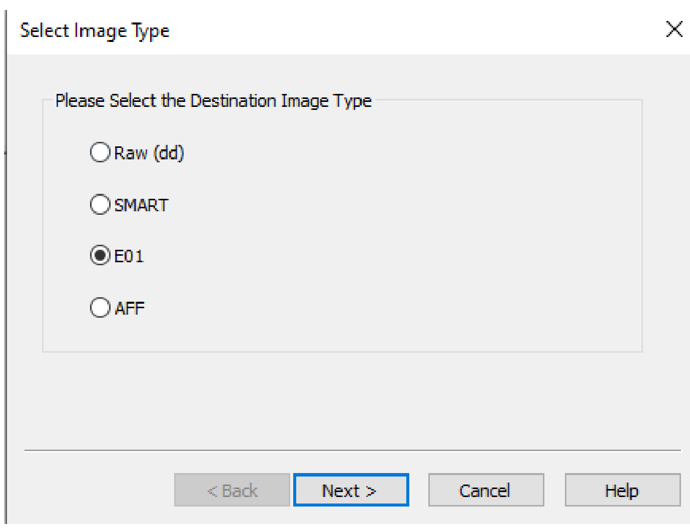

Figure 2.2: Image type E01

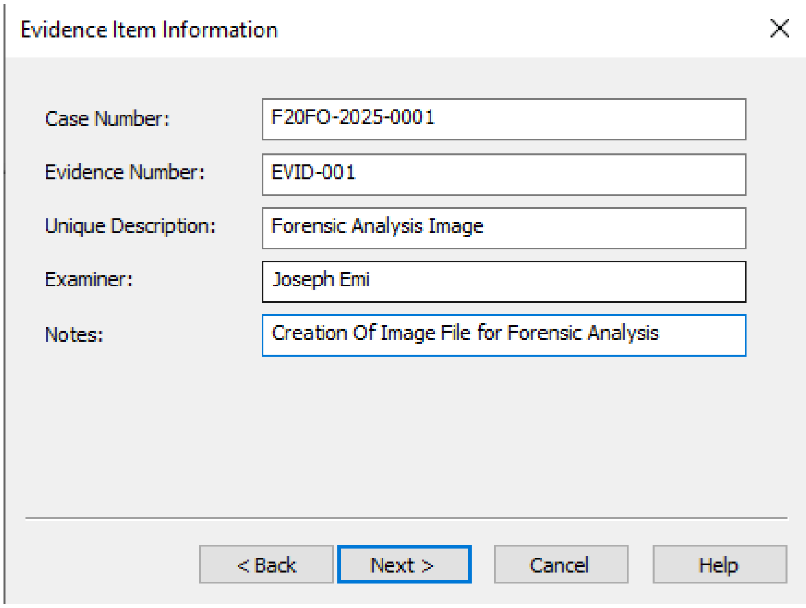

Figure 2.3: FTK image Description

```
 COMPUTED HASH :  1c75bbc9b4314ddbc8893033e732dc3e
 COMPUTED HASH :  7863672aa9dd48213acb0b36fbd6e9dd1fa8bcbf

Image Verification Results:
 Verification started:  Thu Apr  3 14:26:08 2025
 Verification finished: Thu Apr  3 14:28:22 2025
 MD5 checksum:     1c75bbc9b4314ddbc8893033e732dc3e : verified
 SHA1 checksum:    7863672aa9dd48213acb0b36fbd6e9dd1fa8bcbf : verified
```

Figure 2.4: Image Hash Values for FTK

Name	Date modified	Type	Size
Image.E01	03/04/2025 2:24 PM	E01 File	1,535,892 KB
Image.E01	03/04/2025 2:28 PM	Text Document	2 KB
Image.E02	03/04/2025 2:25 PM	E02 File	1,535,910 KB
Image.E03	03/04/2025 2:25 PM	E03 File	1,535,875 KB
Image.E04	03/04/2025 2:25 PM	E04 File	1,535,825 KB
Image.E05	03/04/2025 2:26 PM	E05 File	1,113,108 KB

Figure 2.5: Images from FTK In folder

Figure 2.6: Creating Mount for GuyMager

Figure 2.7: Mounting VDMK images onto Ubuntu

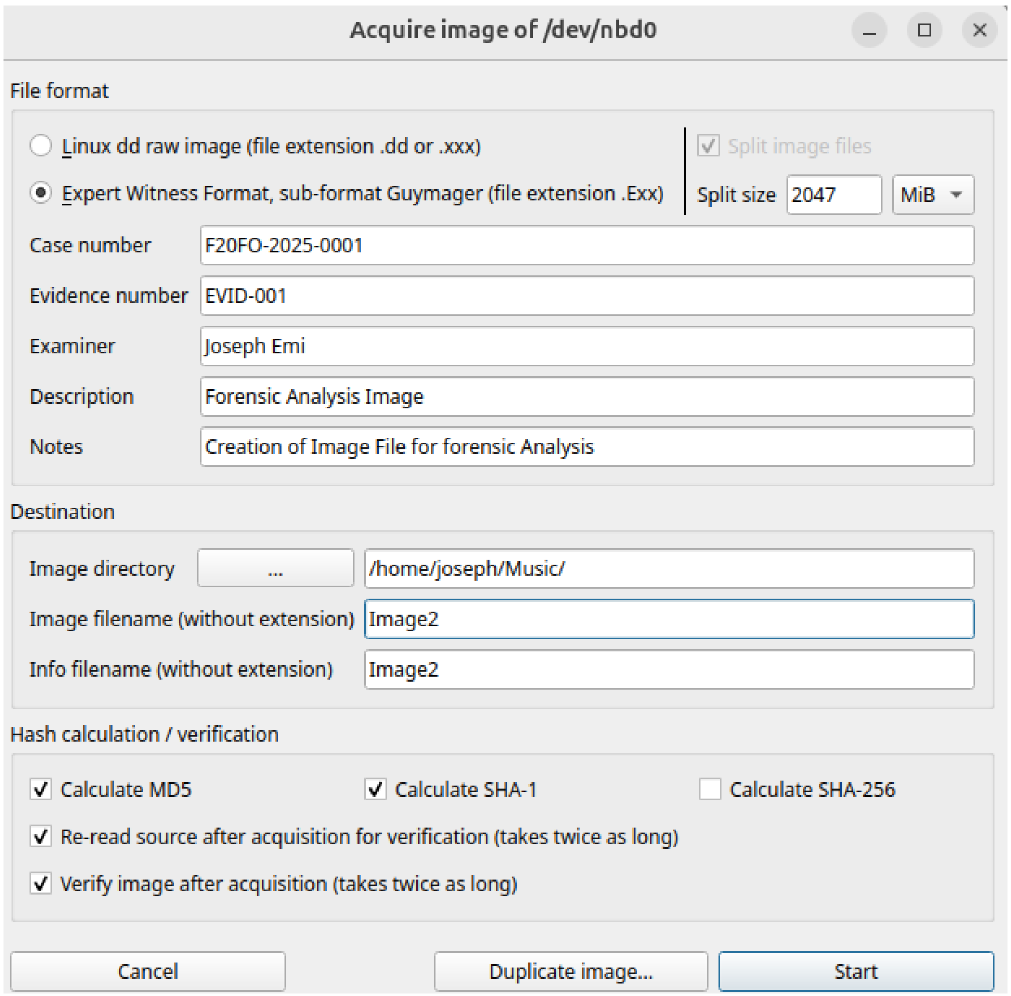

Figure 2.5: Description and Selection of Image type

Figure 2.8: shows the hash generated from Image using GuyMager

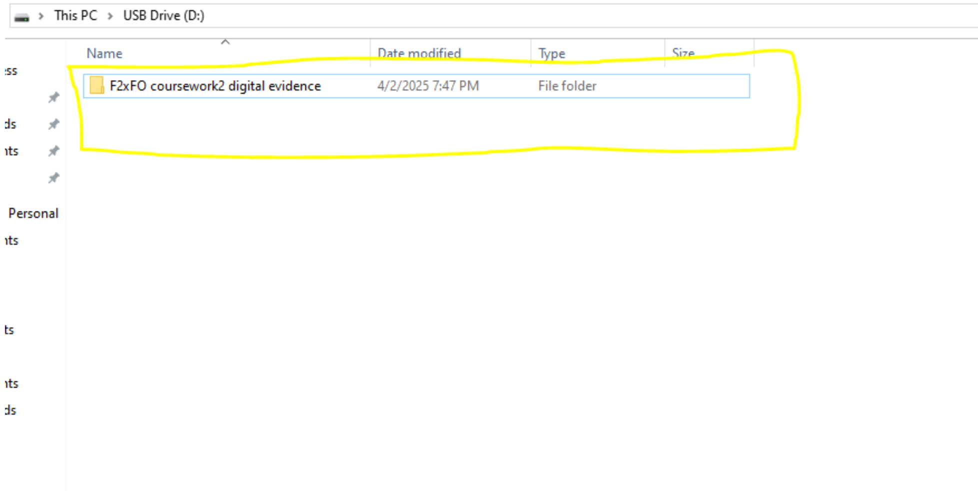

Figure 2.9: USB used to transfer the evidence to VMs

Appendix B: Evidence Pictures

Tag	File	Comment	User Name	Modified Time	Changed Time	Accessed Time	Created Time	Size (Bytes)
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/.bash_history		Joseph	2020-03-15 07:43:23 GMT-07:00	2020-03-15 07:43:23 GMT-07:00	2022-03-07 03:34:02 GMT-08:00	2020-03-15 03:38:22 GMT-07:00	409
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/my-ssh		Joseph	2020-03-15 03:22:15 GMT-07:00	2020-03-15 03:22:15 GMT-07:00	2022-03-07 03:34:02 GMT-08:00	2020-03-15 03:22:15 GMT-07:00	3326
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Pictures/bolivia.png		Joseph	2020-03-15 10:02:20 GMT-07:00	2020-03-15 06:24:22 GMT-07:00	2020-03-14 17:00:00 GMT-07:00	2020-03-15 06:24:21 GMT-07:00	4802048
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/one day in heaven		Joseph	2019-08-12 15:59:16 GMT-07:00	2020-03-15 07:00:19 GMT-07:00	2022-03-07 03:35:19 GMT-08:00	2020-03-15 06:55:47 GMT-07:00	26993148
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/myhiddenfile.txt		Joseph	2012-10-23 11:07:11 GMT-07:00	2020-03-15 07:12:50 GMT-07:00	2020-03-15 07:12:51 GMT-07:00	2020-03-15 07:12:50 GMT-07:00	2012724
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/sys_cooking.dll		Joseph	2020-03-15 06:48:15 GMT-07:00	2020-03-15 06:57:02 GMT-07:00	2020-03-15 06:56:26 GMT-07:00	2020-03-15 06:55:47 GMT-07:00	1708444
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/sys_phare.dll		Joseph	2020-03-15 06:49:57 GMT-07:00	2020-03-15 06:57:02 GMT-07:00	2020-03-15 06:51:42 GMT-07:00	2020-03-15 06:55:47 GMT-07:00	813706
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/3720-Computer Forensic Communication 2001-3.pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:44 GMT-07:00	2022-03-07 03:34:31 GMT-08:00	2020-03-15 06:23:20 GMT-07:00	383535
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/kanban		Joseph	2019-08-12 15:59:02 GMT-07:00	2020-03-15 06:59:29 GMT-07:00	2020-03-15 06:59:29 GMT-07:00	2020-03-15 06:55:47 GMT-07:00	55209
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/3746-GENERAL PROCEDURES for using EnCase.pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:44 GMT-07:00	2020-03-15 06:30:34 GMT-07:00	2020-03-15 06:23:20 GMT-07:00	52748
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/3730-Computer_examination_methods.pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:46 GMT-07:00	2020-03-15 06:30:34 GMT-07:00	2020-03-15 06:23:20 GMT-07:00	16137
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/digital forensics/3720-Computer Forensic Communication 2001-3.pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:44 GMT-07:00	2022-03-07 03:34:31 GMT-08:00	2020-03-15 06:23:20 GMT-07:00	383535
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/digital forensics/3717-Computer Forensic Communication 2001-1.pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:44 GMT-07:00	2020-03-15 06:30:34 GMT-07:00	2020-03-15 06:23:20 GMT-07:00	283353
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/digital forensics/3713-Adequacy of Criminal Law and Procedure (CYBER).pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:44 GMT-07:00	2020-03-15 06:30:33 GMT-07:00	2020-03-15 06:23:20 GMT-07:00	173913
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/digital forensics/3746-GENERAL PROCEDURES for using EnCase.pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:44 GMT-07:00	2020-03-15 06:30:34 GMT-07:00	2020-03-15 06:23:20 GMT-07:00	52748
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/digital forensics/3730-Computer_examination_methods.pdf		Joseph	2020-03-15 09:41:40 GMT-07:00	2020-03-15 06:23:46 GMT-07:00	2020-03-15 06:30:34 GMT-07:00	2020-03-15 06:23:20 GMT-07:00	16137
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Downloads/slacker.exe		Joseph	2012-10-16 06:42:33 GMT-07:00	2020-03-15 07:12:50 GMT-07:00	2019-07-17 12:09:36 GMT-07:00	2020-03-15 07:12:50 GMT-07:00	2804426
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Downloads/timestomp.exe		Joseph	2012-10-23 07:58:31 GMT-07:00	2020-03-15 07:12:50 GMT-07:00	2019-07-17 12:09:36 GMT-07:00	2020-03-15 07:12:50 GMT-07:00	2387303
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Videos/one day in heaven		Joseph	2019-08-12 15:59:16 GMT-07:00	2020-03-15 07:00:19 GMT-07:00	2022-03-07 03:35:19 GMT-08:00	2020-03-15 06:55:47 GMT-07:00	26993148
Notable Item (Notable)	/img_Image.E01/vol_vol2/etc/shadow		Joseph	2020-03-15 06:51:44 GMT-07:00	2020-03-15 06:51:44 GMT-07:00	2022-03-07 03:32:34 GMT-08:00	2020-03-15 06:51:44 GMT-07:00	1620
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/auth.log.2.gz		Joseph	2020-03-15 02:32:42 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	2020-03-09 06:21:01 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	2028
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/syslog.3.gz		Joseph	2020-03-10 13:08:08 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	2020-03-09 13:06:08 GMT-07:00	2020-03-15 02:36:41 GMT-07:00	13373
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/syslog.4.gz		Joseph	2020-03-09 13:06:08 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	2020-03-09 06:21:01 GMT-07:00	2020-03-10 13:08:08 GMT-07:00	34100
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/syslog.2.gz		Joseph	2020-03-15 02:36:40 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	2020-03-10 13:08:08 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	40833
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/kern.log.2.gz		Joseph	2020-03-15 02:31:45 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	2020-03-09 06:21:01 GMT-07:00	2022-03-07 04:17:56 GMT-08:00	42784
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/auth.log.1		Joseph	2022-03-07 04:17:01 GMT-08:00	2022-03-07 04:17:56 GMT-08:00	2020-03-15 02:36:41 GMT-07:00	2020-03-15 02:36:41 GMT-07:00	48849
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/kern.log		Joseph	2022-03-07 04:36:16 GMT-08:00	2022-03-07 04:36:16 GMT-08:00	2022-03-07 04:17:56 GMT-08:00	2022-03-07 04:17:56 GMT-08:00	156792
Notable Item (Notable)	/img_Image.E01/vol_vol2/var/log/syslog		Joseph	2022-03-07 04:36:25 GMT-08:00	2022-03-07 04:36:25 GMT-08:00	2022-03-07 04:17:56 GMT-08:00	2022-03-07 04:17:56 GMT-08:00	197169
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Pictures/Nikon_CoolPixS710_2_13718.JPG		Joseph	2010-04-30 05:38:26 GMT-07:00	2020-03-15 07:18:09 GMT-07:00	2019-07-28 18:58:46 GMT-07:00	2020-03-15 07:18:09 GMT-07:00	5770699
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Pictures/Nikon_CoolPixS710_2_13717.JPG		Joseph	2010-04-30 05:38:26 GMT-07:00	2020-03-15 07:18:09 GMT-07:00	2019-07-28 18:58:45 GMT-07:00	2020-03-15 07:18:09 GMT-07:00	5601687
Notable Item (Notable)	/img_Image.E01/vol_vol2/lib/udev/.#hwdb.binDayJgL		Joseph	2020-03-15 06:51:30 GMT-07:00	2020-03-15 06:51:30 GMT-07:00	2022-03-07 03:32:33 GMT-08:00	2020-03-15 06:51:30 GMT-07:00	9371823
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/.thumbnails/normal/6cf89bbce75b2a7c3560b86b6a154386.png.N73PI1		Joseph	2022-03-07 03:35:47 GMT-08:00	2022-03-07 03:35:47 GMT-08:00	2022-03-07 03:35:47 GMT-08:00	2022-03-07 03:35:47 GMT-08:00	25047
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/.mozilla/firefox/ggx660ss.default/datareporting/archived/2020-03		Joseph	2022-03-07 03:35:46 GMT-08:00	2022-03-07 03:35:46 GMT-08:00	2022-03-07 03:35:46 GMT-08:00	2022-03-07 03:35:46 GMT-08:00	21380
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/513-Solar System/Comets		Joseph	2020-03-15 06:25:40 GMT-07:00	2020-03-15 06:25:40 GMT-07:00	2020-03-15 06:25:40 GMT-07:00	2020-03-15 06:25:40 GMT-07:00	20183
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/2723-Pulsars/ROG Pulsars_files/crabpurplehirez_m.jpg		Joseph	2020-03-15 09:28:46 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	2020-03-14 17:00:00 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	13083
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/2723-Pulsars/ROG Pulsars_files/crab_vlt.jpg		Joseph	2020-03-15 09:28:46 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	2020-03-14 17:00:00 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	11807
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/2723-Pulsars/ROG Pulsars_files/neutron.jpg		Joseph	2020-03-15 09:28:46 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	2020-03-14 17:00:00 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	9252
Notable Item (Notable)	/img_Image.E01/vol_vol2/home/f21fo-cw2/Documents/2723-Pulsars/ROG Pulsars_files/rogbanner_50px.jpg		Joseph	2020-03-15 09:28:46 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	2020-03-14 17:00:00 GMT-07:00	2020-03-15 06:23:11 GMT-07:00	2878

Figure 2.10: Shows all the notable items that were tagged

```
$oldoffice$1*867d6947deca0b62fe61323eecceef1d*ed7f90d4f425227d38d7e2d191877b0c*0b4acde8f1edf21481f0e4daaf1435be:relativity

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 9700 (MS Office <= 2003 $0/$1, MD5 + RC4)
Hash.Target......: $oldoffice$1*867d6947deca0b62fe61323eecceef1d*ed7f9...1435be
Time.Started.....: Sun Apr 06 20:05:04 2025 (0 secs)
Time.Estimated...: Sun Apr 06 20:05:04 2025 (0 secs)
Kernel.Feature...: Optimized Kernel
Guess.Base.......: File (rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:  3613.8 kH/s (4.97ms) @ Accel:512 Loops:1 Thr:32 Vec:1
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 1050388/14344384 (7.32%)
Rejected.........: 1812/1050388 (0.17%)
Restore.Point....: 787532/14344384 (5.49%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: snuppy1 -> KEKE101
Hardware.Mon.#1..: Temp: 45c Util: 67% Core: 652MHz Mem: 810MHz Bus:8

Started: Sun Apr 06 20:04:55 2025
Stopped: Sun Apr 06 20:05:06 2025

F:\hashcat\hashcat-6.2.6>hashcat -m 9700 -a 0 hash.txt rockyou.txt
hashcat (v6.2.6) starting
```

Figure 2.11: Shows hashcat cracking einsteins doc

```
$oldoffice$1*5ad27311a1f1416b23b270f5b6033dd8*c26b448eeb17dbd37eb912ff514c7a9d*c3fc7d94a1a7a65ef87394921515ff4b:wavelength

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 9700 (MS Office <= 2003 $0/$1, MD5 + RC4)
Hash.Target......: $oldoffice$1*5ad27311a1f1416b23b270f5b6033dd8*c26b4...15ff4b
Time.Started.....: Sun Apr 06 20:09:48 2025 (1 sec)
Time.Estimated...: Sun Apr 06 20:09:49 2025 (0 secs)
Kernel.Feature...: Optimized Kernel
Guess.Base.......: File (rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........: 13909.1 kH/s (3.62ms) @ Accel:1024 Loops:1 Thr:32 Vec:1
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 1577729/14344384 (11.00%)
Rejected.........: 4865/1577729 (0.31%)
Restore.Point....: 1050388/14344384 (7.32%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: KEKE -> lilangels2
Hardware.Mon.#1..: Temp: 46c Util: 30% Core:1500MHz Mem:6000MHz Bus:8

Started: Sun Apr 06 20:09:47 2025
Stopped: Sun Apr 06 20:09:50 2025

F:\hashcat\hashcat-6.2.6>hashcat -m 9700 -a 0 hash.txt rockyou.txt
hashcat (v6.2.6) starting
```

Figure 2.12: shows hashcat cracking Nicholas doc


```
$oldoffice$1*5ad27311a1f1416b23b270f5b6033dd8*c26b448eeb17dbd37eb912ff514c7a9d*c3fc7d94a1a7a65ef87394921515ff4b:wavelength

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 9700 (MS Office <= 2003 $0/$1, MD5 + RC4)
Hash.Target......: $oldoffice$1*5ad27311a1f1416b23b270f5b6033dd8*c26b4...15ff4b
Time.Started.....: Sun Apr 06 20:09:48 2025 (1 sec)
Time.Estimated...: Sun Apr 06 20:09:49 2025 (0 secs)
Kernel.Feature...: Optimized Kernel
Guess.Base.......: File (rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........: 13909.1 kH/s (3.62ms) @ Accel:1024 Loops:1 Thr:32 Vec:1
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 1577729/14344384 (11.00%)
Rejected.........: 4865/1577729 (0.31%)
Restore.Point....: 1050388/14344384 (7.32%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: KEKE -> lilangels2
Hardware.Mon.#1..: Temp: 46c Util: 30% Core:1500MHz Mem:6000MHz Bus:8

Started: Sun Apr 06 20:09:47 2025
Stopped: Sun Apr 06 20:09:50 2025

F:\hashcat\hashcat-6.2.6>hashcat -m 9700 -a 0 hash.txt rockyou.txt
hashcat (v6.2.6) starting
```


Figure 2.13: shows hashcat cracking another doc

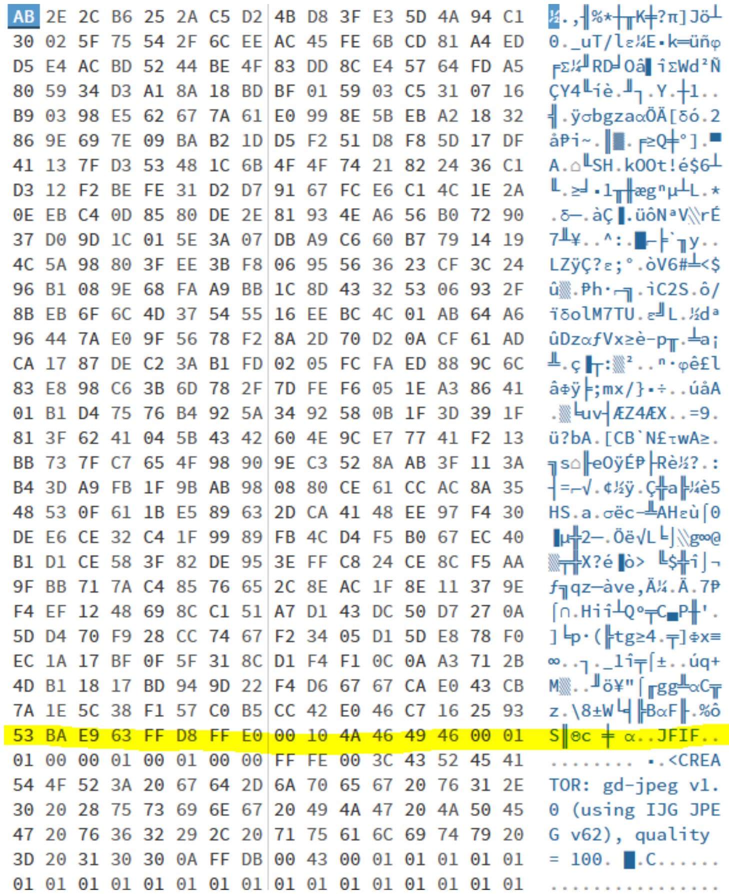

Figure 2.14: shows HexEdit where there is obfuscation, and the real image starts from highlight

Figure 2.15: Image carved after removing obfuscation

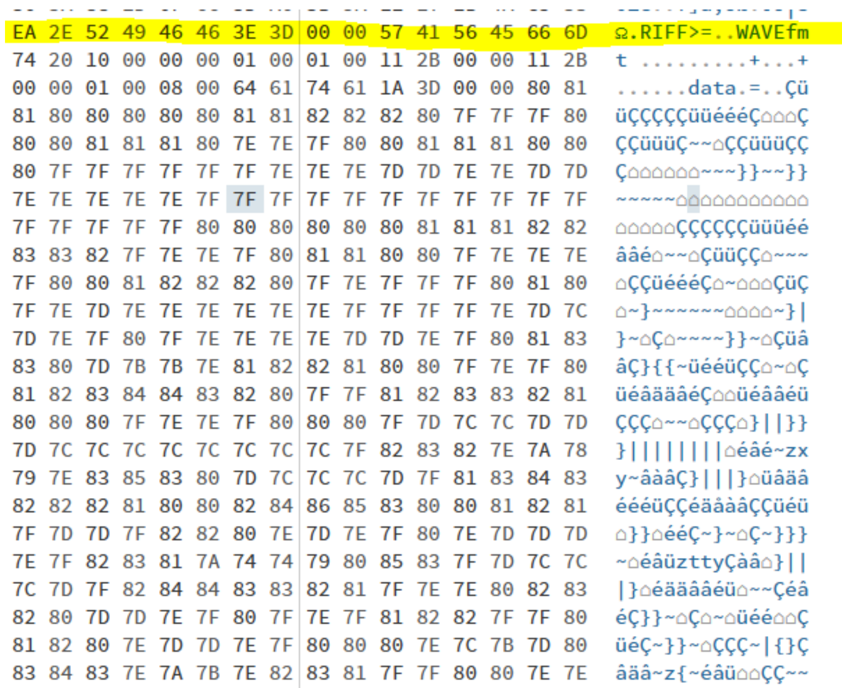

Figure 2.16: Carved image has embedded Wav file

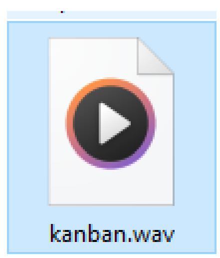

Figure 2.17: Carved image has embedded Wav file which was extracted

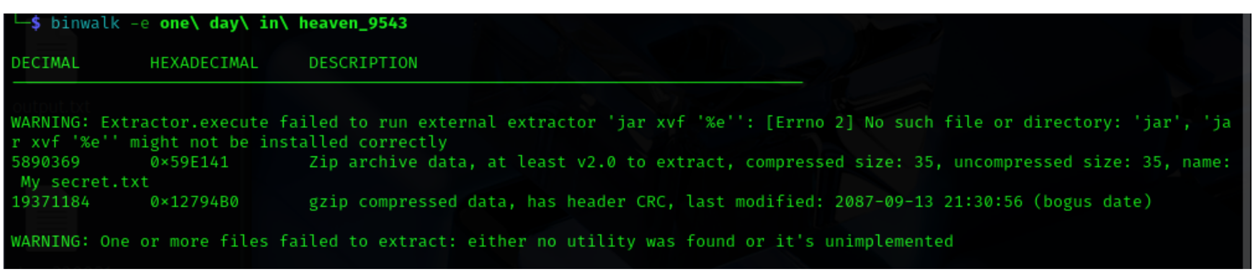

Figure 2.18: binwalk extraction on "one day in heaven" file

Figure 2.19: Carved images from “one day in heaven” file

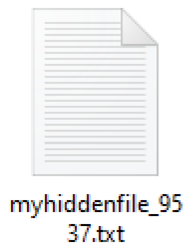

Figure 2.20: File “myhiddenfile” has an extension mismatch

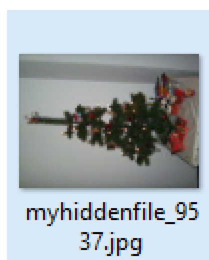

Figure 2.21: File “myhiddenfile” extension changes to JPEG

```
f21fo-cw2:$6$ErLwHYsT$XeRlRufkTGWHq6jMMCn4U6YffQstjsW/Q9BiAiXSyTsbgL4F.98Rq9W8OUvlJF3x9No/UgFXuPG0Z8JMap.iB1:18330:0:99999:7:::
student1:$6$tvNGNUdk$lXtW93OYxAxCXlY4gTJ.9h6VaZM8IvvZrSRRMjZPQRnNOETAZontTL1ckTOjR1bWt61lOy1w2zKmDGyU62Dg.0:18336:0:99999:7:::
student2:$6$9y77CbPE$O2Ee/kiev7.wri6cktZxyAQ4LYUOgeBK6gXIc7DB2WP02VOYKTjFQS.mqatTDwMuxN6bEXDd2Xue0Jpl6drVw1:18336:0:99999:7:::
tutor1:$6$/mNtkM8W$0rAkJLS.mh3U8/m4dmMU07WlSE6SYbKvQiaYvzA55oPD89j8BLEz35nn2UK2E9lwRsvXVL95KL3RgREHg1i6E1:18336:0:99999:7:::
user_one:$6$CrM8lUFK$IVhhHYc0SBpOEbYWqC5KOIPtv30cJPTed771wzhh2nEr6i.4CztMBijOBxT9eir1KTpYYz9F/f1r6hteDWMaH0:18336:0:99999:7:::
user_ten:$6$0u4Fh7n2$p.jkx2twc57f8PHqVQxaNZzxCM/AAp/91jNPfYlI1VejjlTypJW4wfh3Q4/8FcHYBM6YpTYlNILu8Z0zkLQq//:18336:0:99999:7:::
saned:*:18336:0:99999:7:::
```

Figure 2.22: Shadow file from the image exposing users hashes which can be used to brute force later (if possible)

Appendix C: Lab-Setup

The virtual lab was created using VMware Workstation Pro, where both Ubuntu and Windows operating systems were installed on separate virtual machines (VMs). The VMs are configured to operate in isolation, meaning they can only communicate internally and have no access to any external

- Evidence Integrity: By preventing external communication, the virtual machines ensure that the forensic evidence remains uncontaminated and secure from external interference, preserving the authenticity and admissibility of evidence during an investigation.

- Controlled Environment: This isolated configuration also provides complete control over the environment, which allows for precise testing, monitoring, and evaluation of forensic tools and procedures without external variables affecting the results.

Below are screenshots of the VM configurations, which show their isolation.

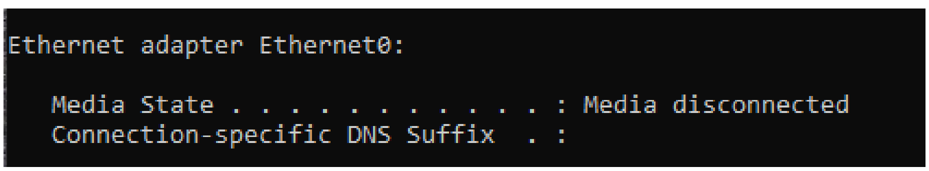

Figure 2.23: Windows IP configuration

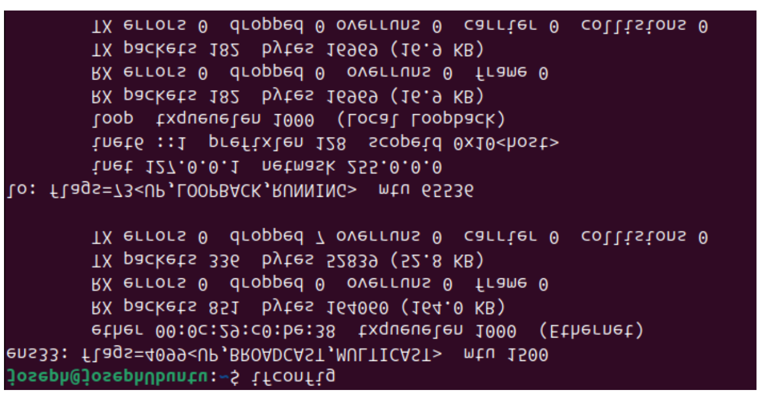

Figure 2.24: Ubuntu IP configuration

Appendix D: Chain of Custody

F20FO Department of Forensic Analysis

EVIDENCE CHAIN OF CUSTODY TRACKING FORM

Case Number: 2025-DF-001 Offense: Unauthorized Access to Virtual Machine

Submitting Officer: Joseph Emmanuel [redacted]

Victim: [redacted]

Suspect: [redacted]

Date/Time Seized: 2025-04-03 12:00 Location of Seizure: [redacted]

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
1	1	VM files (VMDK) suspected of unauthorized access: digital forensic image (E01 Format) created using FTK imager: Associated Hash: (SHA1): 7863672aa9dd48213acb0b36fbd6e9dd1fa8bcbf: (MD5): 1c75bbc9b4314ddbc8893033e732dc3e

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
1	2025-04-03 12:15	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Evidence collected from given folder within canvas and downloaded to PC hard drive

1	2025-04-03 12:45	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Evidence files (VM disks) have been transferred from PC hard drive to USB
1	2025-04-03 13:40	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Evidence Files (VM Disks) Transferred from USB to Windows VM for imaging process. Evidence transferred to a separate logical section of the VM hard drive to prevent contamination. VM is also isolated, and there is no connection or shared folders initiated.
1	2025-04-03 14:10	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Evidence used on FTK imager to capture a forensic image in the E01 format. Computed hashes: (SHA1): 7863672aa9dd48213acb0b36fbd6e9dd1fa8bcbf: (MD5): 1c75bbc9b4314ddbc8893033e732dc3e
1	2025-04-03 16:24	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Forensic Image loaded onto Autopsy within Windows with case number: F20F0-2025-0001. Case Name: Virtual Machine Investigation – Unauthorized VM Use

1	2025-04-04 13:21	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Autopsy was used to analyze the images files system to look for suspicious files
1	2025-04-05 14:09	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	During the analyses certain interesting files were found on the suspect's disk: Bash_history: 8be22334ac2ce64394bfe2338ba73bf4 my-ssh: 8b38f341d4f8f5174fc246ea6fbcd837 one day in heaven: 7bccf5ea225501db9b6c429c9c3e0c6a myhiddenfile.txt: 621746a5e5ebc6ea78668c9bc0077862 sys_cooking.dll: 27eaf60fc802b9526cb16483911e5e03 sys_phare.dll: ef54b32bf0c8afd0c736ed7000f2b77f kanban: 43e534fbf06186c3c9aec1e83f805ec9 slacker.exe: f8dd82ed9d63b57

				ccad05dc1dae333cb These files were tagged within autopsy, and a report was generated revealing its hashes
1	2025-04-06 10:02	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Further analysis was conducted the next day and more notable files were discovered: timestomp.exe: 817f3508132c5c0cf24faa3428332a2c shadow: 032e651cf16ffaddc63efeee1d5e6f26 kern.log: 97ca6a6cdb1e7aa352bbbe7b14b296eb syslog: 9f205046e4c7b526294242c456434496 2761-Hewish Work[8384].doc: 3a53c2d9289f32c371a9fa62d14ee16d

				X3fw.ncf: 8ed8fbdb4c4862133c233c88c34e4171 Agfa_DC-733s_0_936.JPG: cdec2617afc3750c46ee6792b5fdbf93 4074-Nicolaus Copernicus.doc: 7f11b6d39e68e14a5ca36340f8bd6398 Albert Einstein.doc: 8ed8fbdb4c4862133c233c88c34e4171 These files were tagged within autopsy, and a report was generated revealing its hashes
1	2025-04-06 15:14	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	All the evidence was then extracted using report generation where the notable evidence was put into a separate folder for further deeper analysis
1	2025-04-06 15:30	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Certain files were detected by

				Autopsy to have extension mismatch. These files were: myhiddenfile.txt: 621746a5e5ebc6ea78668c9bc0077862 sys_cooking.dll: 27eaf60fc802b9526cb16483911e5e03 sys_phare.dll: ef54b32bf0c8afd0c736ed7000f2b77f slacker.exe: f8dd82ed9d63b57ccad05dc1dae333cb timestomp.exe: 817f3508132c5c0cf24faa3428332a2c Agfa_DC-733s_0_936.JPG: cdec2617afc3750c46ee6792b5fdbf93 These files all had a mismatched extension of .JPG which was reported by autopsy except for Agfa_DC-733s_0_936.JPG which had a .WAV mismatch

				This was also confirmed by the fact that during the HexEdit analysis the headers of all the files were different from their extension
1	2025-04-06 18:30	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	During an inspection of the seized extracted doc files, the files seem to be encrypted with a password and were unable to open and view its contents. After looking through the disk one more time to no avail I decided to brute force it by converting each .doc file to a hash using john the ripper and using a wordlist rockyou.txt I used hashcat to break into the encryption and retrieved the password for the files. These files were: 4074-Nicolaus Copernicus.doc: 7f11b6d39e68e1

				4a5ca36340f8bd6398 Albert Einstein.doc: 8ed8fbdb4c4862133c233c88c34e4171 2761-Hewish Work[8384].doc: 3a53c2d9289f32c371a9fa62d14ee16d However one file did not get brute forced due to unrecognizable hash format which was: X3fw.ncf: 8ed8fbdb4c4862133c233c88c34e4171
1	2025-04-06 19:23	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	The last set of files were more obfuscated than the last as extraction of information from those files showed barely anything. Using Binwalk I was able to extract a text file from on the files and that's about it. However, further inspection of

				those files using HexEdit I was able to notice that there are many embedded images and WAV formats within a single file and hence why the large file size. Using Hexedit I was able to carve out specific places within the file and extract specific images which showed a lot of relevant information. The files that had this embedded information were: one day in heaven: 7bccf5ea225501db9b6c429c9c3e0c6a kanban: 43e534fbf06186c3c9aec1e83f805ec9
1	2025-04-06 19:23	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	All this evidence has been noted within the extensive report that is done along with this document.

1	2025-04-07 09:34	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Evidence File was transferred from the USB to the Ubuntu VM. These VMDK files will be used to create an image using GuyMager within Ubuntu
1	2025-04-07 10:11	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	Using qemu the disk was mounted onto the virtual machine and ready for imaging
	2025-04-07 10:18	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	The disk was detected by GuyMager and had gone through a similar process like FTK imager to create the image. The E01 format was chosen to create the image with similar descriptions.
	2025-04-07 10:33	Joseph Emmanuel [redacted]	Joseph Emmanuel [redacted]	The image was created and the hash for the images were generated: MD5 hash : 1c75bbc9b4314d dbc8893033e732 dc3e MD5 hash verified source : 1c75bbc9b4314d

				dbc8893033e732 dc3e MD5 hash verified image : 1c75bbc9b4314d dbc8893033e732 dc3e SHA1 hash : 7863672aa9dd48 213acb0b36fbd6 e9dd1fa8bcbf SHA1 hash verified source : 7863672aa9dd48 213acb0b36fbd6 e9dd1fa8bcbf SHA1 hash verified image : 7863672aa9dd48 213acb0b36fbd6 e9dd1fa8bcbf

Final Disposal Authority

Authorization for Disposal

Item(s) #: 18 on this document pertaining to (suspect): [redacted]
is(are) no longer needed as evidence and is/are authorized for disposal by (check appropriate disposal method)

☐ Return to Owner ☐ Auction/Destroy/Divert

Name & ID# of Authorizing Officer: Joseph Emmanuel Signature: Joseph Date: 07/04/2025

Witness to Destruction of Evidence

Item(s) #: __________ on this document were destroyed by Evidence Custodian
________________________ID#:______
in my presence on (date) __________________________.
Name & ID# of Witness to destruction: ________________________ Signature: ______________________Date: _______________

Release to Lawful Owner

Item(s) #: __________ on this document was/were released by Evidence Custodian
________________________ID#:_________ to
Name ____
Address: ____ City: ____________________State: _______ Zip Code: __________
Telephone Number: (_____) ___________________________________
Under penalty of law, I certify that I am the lawful owner of the above item(s).

Signature: ____ Date: _________________________

Copy of Government-issued photo identification is attached. ☐ Yes ☐ No

This Evidence Chain-of-Custody form is to be retained as a permanent record by the Anywhere Police Department.